

ABSTRACT

Industrial zones are very prone to accidental gas leak which could cause casualties, damage to equipment and severe hazard to the environment. Conventional monitoring approaches based on fixed-threshold comparisons are shown to be weak in reliability in noisy/dynamic environments and incapable of providing an estimated spatial origin for a leak. This paper provides an IoT-based distributed low-cost safety monitoring system for chemical factories based on probabilistic leak detection with spatial localisation. The sensor is selected carefully to cover temperature, humidity, and air quality to coarsely measure the well-being of the environment (air quality). The proposed architecture uses wireless communication through the ESP-NOW protocol between three ESP32-based sensor nodes, each containing a gas sensor (MQ-135) and a temperature–humidity sensor (DHT11), with a master node, and Bayesian inference is performed on each node to derive a leak confidence score, which is a probabilistic interpretation of raw sensor readings, to avoid false alarms in the presence of environmental disturbances. A Weighted Centroid Localisation (WCL) algorithm on the master node combines per-node probability scores to physically localise the leak. Alerts are designated as Safe, Warning, or Danger, and streaming of sensory data to the ThingSpeak IoT cloud visualisation platform is initiated. System evaluation in a controlled indoor environment demonstrates multi-node gas detection, spatial estimation, and probabilistic filtering with reliability.

Keywords: IoT; ESP32; ESP-NOW; Gas Leak Detection; Bayesian Inference; Weighted Centroid Localisation; Industrial Safety; MQ135; Chemical Factory Monitoring

Introduction

Industrial environments such as chemical plants, oil refineries, and manufacturing units involve the handling of hazardous materials and gases, making safety a critical concern. Traditional safety systems are often limited to manual inspection or isolated monitoring devices, which may not provide timely detection of dangerous conditions. This creates a need for more advanced and reliable safety solutions.

The **Internet of Things (IoT)** has emerged as a powerful technology for improving industrial safety systems. IoT refers to a network of interconnected devices equipped with sensors, microcontrollers, and communication modules that can collect and exchange data in real time. In industrial safety applications, IoT enables continuous monitoring of environmental parameters such as gas concentration, temperature, humidity, and pressure.

An IoT-based industrial safety system typically consists of distributed sensor nodes placed at different locations within the industrial area. These nodes collect data from sensors and transmit it to a central processing unit or cloud platform. The system analyzes the data to detect abnormal conditions and generates alerts when necessary. This real-time monitoring capability significantly reduces response time and helps prevent accidents.

One of the key advantages of IoT-based systems is their ability to provide **remote monitoring and control**. Users can access system data through dashboards or mobile applications, allowing them to monitor conditions from anywhere. Additionally, IoT systems can store historical data, which can be used for analysis and improving safety measures.

Modern IoT-based safety systems also incorporate intelligent techniques for data analysis. By using probabilistic or data-driven approaches, these systems can estimate the severity of a hazard and reduce false alarms. Furthermore, wireless communication technologies enable efficient data transfer without the need for complex wiring, making the system more flexible and scalable.

Overall, IoT-based industrial safety systems offer a smart, efficient, and cost-effective solution for enhancing safety in hazardous environments. They improve detection accuracy, enable faster response, and provide better situational awareness compared to traditional safety systems.

1.1. Overview of Industrial Safety in Chemical Factories.

Chemical factories are among the most sensitive industrial environments due to the presence of hazardous substances, toxic gases, and highly reactive chemicals. These industries involve processes such as chemical reactions, storage of volatile materials, and transportation of gases, all of which pose significant safety risks if not properly monitored and controlled.

Industrial safety in chemical factories focuses on preventing accidents, protecting human life, minimizing environmental damage, and ensuring smooth operation of industrial processes. Common hazards in such environments include gas leaks, fire outbreaks, explosions, chemical spills, and exposure to toxic substances. Among these, gas leakage is one of the most critical risks, as many gases are difficult to detect without proper monitoring systems (**Wang et al., 2022 [1]**).

To manage these risks, chemical industries implement various safety measures such as gas detection systems, ventilation systems, emergency shutdown mechanisms, and safety protocols. IoT-based monitoring systems have been increasingly used to improve safety by enabling real-time detection and alerts (**Manjunathan et al., 2023 [2]**; **Gnoni et al., 2020 [6]**; **Babu et al., 2024 [18]**). These systems use sensors and cloud platforms to continuously monitor environmental conditions. Wireless Sensor Networks (WSNs) are also widely applied for industrial safety monitoring. They allow deployment of multiple sensor nodes for better area coverage and early detection of gas leaks (**Somov et al., 2011 [3]**; **Guo et al., 2019 [5]**). However, challenges such as power consumption and system complexity still exist.

Advanced techniques such as machine learning and probabilistic models are being used to improve detection accuracy and reduce false alarms. Decision tree models and deep learning approaches have shown promising results in gas leak detection (**Yuan et al., 2025 [17]**; **Zhang et al., 2023 [11]**). Bayesian-based methods further enhance reliability by incorporating probabilistic reasoning (**Guerriero et al., 2016 [8]**; **Carpenter et al., 2006 [12]**).

Sensor accuracy and reliability are also important factors in industrial safety systems. Calibration techniques and drift compensation methods are used to maintain long-term accuracy of sensors (**Fernandez et al., 2016 [22]**; **Haugen et al., 2000 [23]**). Multi-sensor data fusion techniques further improve system performance by combining data from different sources (**Fu et al., 2009 [47]**).

Modern communication technologies such as ESP32-based systems using ESP-NOW protocol provide fast and reliable data transmission without requiring internet infrastructure (**Pasic et al., 2021 [42]**; **Mutiara et al., 2024 [43]**). Additionally, IoT platforms enable remote monitoring and visualization of system data (**Sindhwani et al., 2022 [48]**; **Maulana et al., 2022 [49]**).

Despite these advancements, many existing systems still lack features such as accurate leak localization, integrated analysis, and efficient real-time communication. Therefore, there is a need for advanced systems that combine sensing, communication, and intelligent processing to improve industrial safety.

1.2. Challenges in Gas Leak Detection and Monitoring.

Gas leak detection and monitoring in industrial environments is a complex task due to various technical and environmental challenges. One of the primary challenges is the **behavior of gas diffusion**, which depends on multiple factors such as wind speed, temperature, pressure, and atmospheric conditions. These variations make it difficult to accurately predict the spread and concentration of leaked gases in real time (Wang et al., 2022 [1]).

Another major challenge is the **limited accuracy and sensitivity of gas sensors**. Many low-cost sensors are affected by environmental conditions and may produce noisy or drifted readings over time. Maintaining sensor accuracy requires proper calibration and compensation techniques, which increases system complexity (Fernandez et al., 2016 [22]; Haugen et al., 2000 [23]).

Coverage and scalability are also significant issues in gas monitoring systems. Deploying a sufficient number of sensors to cover large industrial areas can be costly and difficult to manage. While Wireless Sensor Networks (WSNs) improve coverage, they introduce challenges such as energy consumption, communication reliability, and network maintenance (Somov et al., 2011 [3]; Guo et al., 2019 [5]).

Another important challenge is the **lack of accurate leak localization**. Many systems can detect the presence of gas but cannot determine the exact or approximate location of the leak. This delays response time and reduces the effectiveness of safety measures (Wu et al., 2020 [10]; Zhang et al., 2023 [11]).

The issue of **false alarms and unreliable detection** is also critical. Sensor noise, environmental variations, and improper threshold settings can lead to incorrect alerts. Advanced techniques such as probabilistic models and data fusion are required to improve reliability and reduce false alarms (Guerriero et al., 2016 [8]; Carpenter et al., 2006 [12]).

In addition, **data processing and analysis** pose challenges, especially when dealing with large amounts of sensor data. Machine learning and intelligent algorithms can improve detection accuracy, but they often require sufficient training data and computational resources (Chen et al., 2018 [4]; Yuan et al., 2025 [17]).

Communication reliability is another concern in industrial environments. Many IoT-based systems depend on Wi-Fi or internet connectivity, which may not always be stable. Efficient communication protocols are required to ensure low latency and reliable data transfer (Pasic et al., 2021 [42]; Mutiara et al., 2024 [43]).

Finally, integrating all these components into a **single efficient system** remains a challenge. Most existing solutions focus on individual aspects such as detection, communication, or analysis, but lack a unified approach that combines real-time monitoring, intelligent decision-making, and localization.

1.3. IoT Applications in Industrial Safety.

The Internet of Things (IoT) has significantly transformed industrial safety systems by enabling real-time monitoring, data collection, and intelligent decision-making. In industrial environments such as chemical factories, IoT-based systems are widely used to detect hazardous conditions, monitor equipment, and ensure worker safety.

One of the primary applications of IoT in industrial safety is **gas leakage detection and monitoring**. IoT-enabled systems use gas sensors and microcontrollers to continuously monitor gas concentration levels and generate alerts when abnormal conditions are detected. These systems provide real-time monitoring and improve response time compared to traditional methods (Manjunathan et al., 2023 [2]; Babu et al., 2024 [18]; Hannan et al., 2018 [20]).

IoT is also used for **environmental monitoring**, where parameters such as temperature, humidity, and air quality are continuously measured. This helps in maintaining safe working conditions and preventing hazardous situations (Gnoni et al., 2020 [6]; Sindhwani et al., 2022 [48]).

Another important application is **wireless sensor networks (WSNs)** for industrial monitoring. IoT-based WSNs allow deployment of multiple sensor nodes across large industrial areas, enabling better coverage and early detection of risks. These systems improve scalability and flexibility in monitoring (Somov et al., 2011 [3]; Guo et al., 2019 [5]).

IoT also supports **remote monitoring and control**, allowing users to access system data through cloud platforms and dashboards. This enables continuous supervision of industrial processes and quick response to abnormal conditions (Maulana et al., 2022 [49]).

Advanced IoT systems incorporate **data analytics and intelligent techniques** to improve safety. Machine learning models are used to analyze sensor data and detect patterns related to potential hazards (Yuan et al., 2025 [17]). Bayesian-based approaches are applied to improve decision-making and reduce false alarms in gas detection systems (Guerriero et al., 2016 [8]; Carpenter et al., 2006 [12]).

In addition, IoT systems use efficient communication protocols to ensure reliable data transmission. Technologies such as ESP32-based communication using ESP-NOW provide low-latency and infrastructure-free communication, making them suitable for industrial environments (Pasic et al., 2021 [42]; Mutiara et al., 2024 [43]).

Overall, IoT applications in industrial safety provide improved monitoring, faster response, better data analysis, and enhanced reliability. These systems play a crucial role in reducing risks, preventing accidents, and ensuring safe operation in industrial environments.

1.4. Need for an IoT-Based Safety Monitoring System.

Industrial environments, especially chemical factories, involve handling hazardous gases and materials that pose serious safety risks. Traditional gas monitoring systems are often limited in functionality, as they rely on manual inspection or standalone sensors with predefined thresholds. These systems lack real-time monitoring, remote accessibility, and intelligent analysis, which reduces their effectiveness in preventing accidents.

The behavior of gas leakage and its diffusion in industrial environments is complex and influenced by several environmental factors, making early detection difficult (**Wang et al., 2022 [1]**). Conventional systems are not capable of adapting to such dynamic conditions, which increases the risk of delayed detection and response.

IoT-based systems have emerged as a solution to overcome these limitations by enabling continuous monitoring and real-time data transmission. Several studies have demonstrated the effectiveness of IoT in gas leakage detection using sensors and cloud platforms (**Manjunathan et al., 2023 [2]**; **Babu et al., 2024 [18]**; **Hannan et al., 2018 [20]**). These systems improve response time and allow remote monitoring of industrial environments.

Wireless Sensor Networks (WSNs) further enhance monitoring capabilities by deploying multiple sensor nodes across a large area. This improves coverage and allows early detection of gas leaks (**Somov et al., 2011 [3]**; **Guo et al., 2019 [5]**). However, issues such as energy consumption, communication reliability, and system scalability still exist.

Advanced techniques such as machine learning and probabilistic models have been introduced to improve detection accuracy. These methods analyze sensor data to identify patterns and reduce false alarms (**Yuan et al., 2025 [17]**; **Zhang et al., 2023 [11]**). Bayesian-based approaches are particularly useful for probabilistic reasoning and decision-making in uncertain conditions (**Guerriero et al., 2016 [8]**; **Carpenter et al., 2006 [12]**).

Sensor reliability is another critical factor in industrial safety systems. Calibration and drift compensation techniques are required to maintain long-term accuracy of sensors (**Fernandez et al., 2016 [22]**; **Haugen et al., 2000 [23]**). Multi-sensor data fusion methods further enhance system performance by combining information from different sensors (**Fu et al., 2009 [47]**).

Communication plays a vital role in IoT-based systems. Efficient protocols such as ESP-NOW enable fast and reliable data transfer without requiring internet infrastructure, making them suitable for industrial environments (**Pasic et al., 2021 [42]**; **Mutiara et al., 2024 [43]**). Cloud platforms and dashboards provide remote access and visualization of data, improving monitoring and decision-making (**Sindhwani et al., 2022 [48]**; **Maulana et al., 2022 [49]**).

Despite these advancements, many existing systems lack integration of real-time monitoring, intelligent analysis, and leak localization, which highlights the need for a more comprehensive and efficient safety monitoring system.

2. The Problem Definition

2.1 Problem Statement

Industrial environments such as chemical plants, oil refineries, and manufacturing units involve the handling of hazardous gases that can pose serious risks if leakage occurs. Gas leaks are often difficult to detect in the early stages, especially in large industrial setups where monitoring is limited or not continuous. Traditional systems either rely on manual inspection or isolated sensors, which may not provide real-time alerts or accurate information about the severity and location of the leak.

The main problem addressed in this project is the **lack of an efficient, real-time, and intelligent gas monitoring system** that can detect leaks early, estimate their severity, and provide timely alerts. Existing systems often fail to provide continuous monitoring, have delayed response times, and do not offer any method to identify the approximate location of the leak. This can lead to increased risk of accidents, damage to equipment, environmental pollution, and threats to human safety.

The need for this project arises from the requirement to enhance industrial safety by using modern technologies. A system is required that can continuously monitor environmental parameters, detect abnormal gas levels, and instantly notify users about potential hazards. Such a system should also be capable of analyzing data intelligently to reduce false alarms and improve reliability.

This problem is particularly important in industries where hazardous gases are present, such as chemical factories, gas storage facilities, and processing plants. Early detection and quick response can significantly reduce the impact of accidents. Therefore, developing a smart, IoT-based safety monitoring system is essential to ensure a safer working environment, protect human life, and minimize industrial losses.

2.2 Scope

The scope of this project focuses on the development of an IoT-based system for monitoring environmental conditions and detecting gas leakage in industrial environments. The system is designed to continuously measure parameters such as gas concentration, temperature, and humidity using sensor nodes deployed at different locations within a facility.

The project covers the implementation of a distributed architecture using multiple ESP32-based sensor nodes that communicate with a central master node through ESP-NOW protocol. The system processes the collected data to identify abnormal conditions and estimate the probability of gas leakage. It also includes a mechanism to provide real-time alerts through visual indicators and cloud-based dashboards.

In addition, the project includes basic localization of the gas leak using data from multiple sensor nodes, allowing approximate identification of the affected area. The system also integrates IoT platforms for remote monitoring, enabling users to access real-time data and system status from anywhere.

However, the scope is limited to small-scale implementation and controlled environments. The project does not include industrial-grade calibration, large-scale deployment, or advanced predictive analytics using large datasets. Despite these limitations, the system demonstrates a practical approach to improving industrial safety using modern technologies.

3. Literature Survey

3.1.Review of Literature on Gas Leakage Detection Systems.

This section presents a summary of important research works related to gas leakage detection, IoT-based monitoring systems, and intelligent safety mechanisms. These studies provide a foundation for understanding existing techniques and identifying gaps for improvement.

1. Wang et al. (2022)

Wang et al. studied the behavior of natural gas leakage using the Gaussian plume model. The research analyzes how different environmental factors such as wind speed, pressure, temperature, and atmospheric stability affect gas diffusion. The study highlights the importance of understanding gas spread patterns to take timely preventive actions in industrial environments.

2. Manjunathan et al. (2023)

This work presents an IoT-based gas leakage detection system using ESP8266 and MQ sensors. The system includes a buzzer and mobile application for real-time monitoring and alert generation. Experimental results show that the system can effectively detect gas leakage at varying distances, making it suitable for small-scale applications.

3. Somov et al. (2011)

Somov et al. developed a wireless sensor network (WSN) for combustible gas monitoring. The system uses semiconductor sensors and introduces a pulse heating technique to reduce power consumption. The proposed system enables early detection of gas leakage and demonstrates improved energy efficiency.

4. Chen et al. (2018)

Chen et al. proposed a gas monitoring system using least squares support vector machines (LS-SVM) optimized with swarm intelligence techniques. The system improves classification accuracy for gas-related risks, especially in environments where limited data is available. It provides a reliable method for early warning in coal mine applications.

5. Guo et al. (2019)

This study introduces a gas leakage monitoring system based on mobile wireless sensor networks. The system integrates GPS, GPRS, and high-accuracy gas sensors to provide real-time monitoring over large areas. The results show that the system is reliable and effective for large-scale industrial applications.

6. Gnoni et al. (2020)

Gnoni et al. proposed an IoT-based intelligent safety management system for industrial environments. The system integrates smart devices and digital platforms to monitor hazardous conditions, track equipment status, and improve worker safety. It demonstrates the potential of IoT in enhancing industrial safety management.

7. Guerriero et al. (2016)

This research introduces a Bayesian data fusion approach for gas leak detection in pipelines. By combining data from multiple sensing systems, the proposed method reduces false alarms and improves detection accuracy. The study highlights the importance of probabilistic models in decision-making.

8. Sui et al. (2025)

Sui et al. proposed a method combining Bayesian networks and particle swarm optimization for gas leak detection. The approach focuses on optimal sensor placement and risk assessment. The results show improved efficiency and robustness in detecting leaks in complex environments.

9. Wu et al. (2020)

Wu et al. developed a system that combines computational fluid dynamics (CFD) with Bayesian inference to estimate the location and rate of gas leakage. The method provides accurate predictions and helps in effective emergency response planning.

10. Zhang et al. (2023)

This study proposes a deep probabilistic graph neural network for gas leak detection and localization. The model uses Bayesian inference and graph learning to improve detection accuracy and handle complex sensor relationships. It achieves high performance even with limited labeled data.

3.2. Literature Survey on IoT-Based Safety Monitoring.

SN	Paper	Advantages	Disadvantages	Key Contribution
1	Wang et al. [1]	Accurate modeling	No real-time system	Gas diffusion analysis
2	Manjunathan et al. [2]	Real-time IoT	Limited range	IoT gas detection
3	Somov et al. [3]	Energy efficient	Complex setup	WSN gas monitoring
4	Chen et al. [4]	High accuracy	Needs data	ML-based classification
5	Guo et al. [5]	Large-scale	High cost	Mobile WSN monitoring
6	Gnoni et al. [6]	Smart safety	Complex system	IoT safety management
7	Ganeshan et al. [7]	Multi-parameter	Complex	Smart monitoring system
8	Guerriero et al. [8]	Low false alarm	Complex	Bayesian fusion
9	Sui et al. [9]	Optimized sensors	Complex	Bayesian + PSO

10	Wu et al. [10]	Accurate estimate	High compute	CFD + Bayesian
11	Zhang et al. [11]	High accuracy	Needs big data	GNN detection
12	Carpenter et al. [12]	Good decisions	Complex	Bayesian network
13	Wang et al. [13]	Risk prediction	Complex	Dynamic Bayesian
14	Flores-Cortez et al. [14]	Remote monitoring	Limited scale	IoT LPG monitoring
15	Hawari et al. [15]	Air monitoring	Limited use	Pollution tracking
16	Pandey et al. [16]	Emission study	Less alerts	GHG monitoring
17	Yuan et al. [17]	High ML accuracy	Data dependent	ML leak detection
18	Babu et al. [18]	Smart alerts	Limited range	IoT detection
19	Asthana et al. [19]	Worker safety	Limited features	Sewage gas monitoring
20	Hannan et al. [20]	Low cost	Limited scale	Arduino IoT detection
21	Evalina et al. [21]	Simple system	No IoT	Basic gas detection
22	Fernandez et al. [22]	Accuracy improve	Calibration complex	Sensor calibration
23	Haugen et al. [23]	Stability	Not real-time	Drift compensation
24	Margaritis et al. [24]	Low cost	Needs ML tuning	Calibration method
25	Petersen et al. [25]	Industrial use	Power issue	Wireless instrumentation
26	Kumar et al. [26]	Hazard detection	Low intelligence	IoT monitoring
27	Kavitha et al. [27]	Predictive	Less accuracy	Smart gas system
28	Kodali et al. [28]	ESP32 alerts	Basic analysis	IoT safety system
29	Gura et al. [29]	Multi-gas	Limited processing	Air quality system
30	Maradzhiev et al. [30]	Training model	Not industrial	Educational system

31	Ahmed et al. [31]	High accuracy	Limited scale	LPG detection
32	Kodali et al. [32]	Industrial use	Basic system	Gas safety system
33	Rajitha et al. [33]	GSM alerts	No dashboard	Alert system
34	Ghosh et al. [34]	Multi hazard	Low intelligence	Fire + gas system
35	Siddika et al. [35]	Simple	No automation	LPG detection
36	Tsado et al. [36]	Remote alerts	Low accuracy	GSM system
37	Rakesh et al. [37]	Smart system	Limited optimization	IoT detection
38	Varma et al. [38]	Predictive	Complex	Analytics-based system
39	Chraim et al. [39]	Localization	Low accuracy	Gas localization
40	Murvay et al. [40]	Survey	No implementation	Review study
41	Shu et al. [41]	WSN study	No system	Survey
42	Pasic et al. [42]	Fast comm	Short range	ESP-NOW
43	Mutiara et al. [43]	Performance	No system	QoS analysis
44	Sun et al. [44]	Fault detect	Complex	BRNN method
45	Ng et al. [45]	Fault detect	Limited scale	Bayesian HVAC
46	Verron et al. [46]	Fault isolate	Complex	Bayesian network
47	Fu et al. [47]	Better accuracy	Complex	Sensor fusion
48	Sindhvani et al. [48]	Cloud system	Internet needed	IoT monitoring
49	Maulana et al. [49]	Visualization	Limited detection	Dashboard
50	Kermit et al. [50]	Signal improve	Complex	ICA analysis

3.3. Research Gap in Existing Gas Detection Systems.

Despite significant advancements in gas detection and monitoring technologies, several limitations still exist in current systems. These gaps highlight the need for improved solutions that can provide better accuracy, reliability, and real-time performance in industrial environments.

One of the major gaps in existing systems is the **lack of accurate leak localization**. Many systems are capable of detecting the presence of gas but fail to identify the exact or approximate location of the leak. This limitation delays response time and reduces the effectiveness of safety measures (Wu et al., 2020 [10]; Zhang et al., 2023 [11]; Chraim et al., 2016 [39]).

Another important gap is the **dependence on threshold-based detection methods**. Traditional systems rely on fixed threshold values, which may not adapt well to changing environmental conditions. This often leads to false alarms or missed detections (Manjunathan et al., 2023 [2]; Babu et al., 2024 [18]; Hannan et al., 2018 [20]).

Many existing systems also suffer from **high false alarm rates and lack of intelligent decision-making**. Sensor noise, environmental variations, and improper calibration can affect detection accuracy. Although probabilistic approaches such as Bayesian methods have been proposed, they are not widely implemented in practical IoT systems (Guerriero et al., 2016 [8]; Carpenter et al., 2006 [12]; Sui et al., 2025 [9]).

Another significant gap is related to **sensor reliability and calibration issues**. Gas sensors are prone to drift over time, which reduces their accuracy. Maintaining sensor performance requires calibration techniques that are often complex and not included in many systems (Fernandez et al., 2016 [22]; Haugen et al., 2000 [23]; Margaritis et al., 2018 [24]).

Limited scalability and coverage is also a major concern. While Wireless Sensor Networks (WSNs) improve coverage, they introduce challenges such as energy consumption, network management, and communication reliability (Somov et al., 2011 [3]; Guo et al., 2019 [5]; Petersen et al., 2015 [25]).

Another gap is the **lack of efficient and reliable communication systems**. Many IoT-based solutions depend on Wi-Fi or internet connectivity, which may not be reliable in industrial environments. Although alternative communication methods exist, they are not widely adopted (Pasic et al., 2021 [42]; Mutiara et al., 2024 [43]).

Existing systems also show **limited integration of multiple functionalities**. Most solutions focus only on detection or monitoring, but do not combine detection, localization, and intelligent analysis into a single system (Gnoni et al., 2020 [6]; Ganeshan et al., 2018 [7]; Sindhwani et al., 2022 [48]).

In addition, **data processing and analysis limitations** exist in many systems. While machine learning techniques have been introduced, they often require large datasets and computational resources, making them less suitable for real-time embedded systems (Chen et al., 2018 [4]; Yuan et al., 2025 [17]; Sun et al., 2020 [44]).

Another gap is the **lack of real-time response and remote monitoring capabilities** in traditional systems. Many systems still rely on local alerts and do not provide cloud-based monitoring or visualization features (Maulana et al., 2022 [49]; Rajitha et al., 2017 [33]).

Finally, the **lack of a unified and cost-effective solution** that integrates sensing, communication, localization, and intelligent analysis remains a key challenge. Existing systems either focus on advanced techniques with high complexity or simple systems with limited functionality, but a balanced and practical solution is still required.

4. Proposed System

4.1 Overview

The proposed system is an IoT-based safety monitoring and alert system designed to detect gas leakage and monitor environmental conditions in industrial environments. The system uses multiple sensor nodes placed at different locations to continuously measure parameters such as gas concentration, temperature, and humidity. These sensor nodes are built using ESP32 microcontrollers and communicate with a central master node using ESP-NOW protocol.

The master node collects data from all sensor nodes and processes it to determine the condition of the environment. Based on the sensor readings, the system estimates the probability of gas leakage using a Bayesian-inspired approach. It also identifies the approximate location of the leak using data from multiple nodes. If any abnormal condition is detected, the system generates alerts through LEDs, buzzer, and IoT dashboards for real-time monitoring.

4.1.1. Scope of Implementation

- The system can be implemented in **chemical industries, gas storage units, and manufacturing plants** where gas leakage is a potential risk.
- It supports **real-time monitoring** of environmental parameters using distributed sensor nodes.
- It enables **wireless communication** between nodes without requiring internet connectivity using ESP-NOW.
- It provides **remote monitoring** through cloud platforms such as dashboards for data visualization.
- The system can be expanded by adding more sensor nodes for larger area coverage.

➤ Advantages

- Provides **real-time gas leakage detection**
- Uses **wireless communication (ESP-NOW)** with low latency
- Supports **multi-node monitoring** for better coverage
- Offers **probability-based analysis** for better decision-making
- Includes **basic leak localization capability**
- Cost-effective and easy to implement

➤ Limitations

- Suitable mainly for **small to medium-scale environments**
- Accuracy depends on **sensor calibration and placement**
- Localization provides only **approximate location**, not exact
- Requires periodic maintenance of sensors
- Limited advanced prediction due to absence of large datasets

4.2. Functional modules

The proposed system is divided into several functional modules to simplify design, implementation, and understanding. Each module performs a specific task and works together to achieve the overall objective of real-time gas leakage detection and monitoring.

4.2.1. Sensing Module

This module is responsible for collecting environmental data from the surroundings. It includes gas sensors and temperature-humidity sensors connected to ESP32-based sensor nodes. The sensors continuously measure parameters such as gas concentration, temperature, and humidity and convert them into electrical signals for processing.

4.2.2. Data Acquisition and Processing Module

In this module, the ESP32 microcontroller reads sensor data and performs initial processing such as filtering and formatting. It ensures that the collected data is ready for transmission and further analysis. This module also handles basic validation of sensor readings.

4.2.3. Communication Module

This module manages wireless communication between sensor nodes and the master node using the ESP-NOW protocol. It enables fast, low-latency, and reliable data transfer without requiring a Wi-Fi network. The module ensures that data from multiple nodes is transmitted efficiently to the central system.

4.2.4. Central Processing (Master Node) Module

The master node acts as the central unit of the system. It receives data from all sensor nodes and performs higher-level processing. This includes analyzing sensor readings, estimating gas leak probability using a Bayesian-inspired approach, and determining the overall system status (safe, warning, or danger).

4.2.5. Localization Module

This module estimates the approximate location of the gas leak using data collected from multiple sensor nodes. By comparing sensor readings from different positions, the system identifies the region with higher gas concentration, helping in faster response and mitigation.

4.2.6. Alert and Notification Module

This module generates alerts when abnormal conditions are detected. It activates local indicators such as LEDs and buzzers to warn nearby personnel. It also supports remote alerts through IoT platforms, ensuring timely notification of hazardous situations.

4.2.7. Cloud and Visualization Module

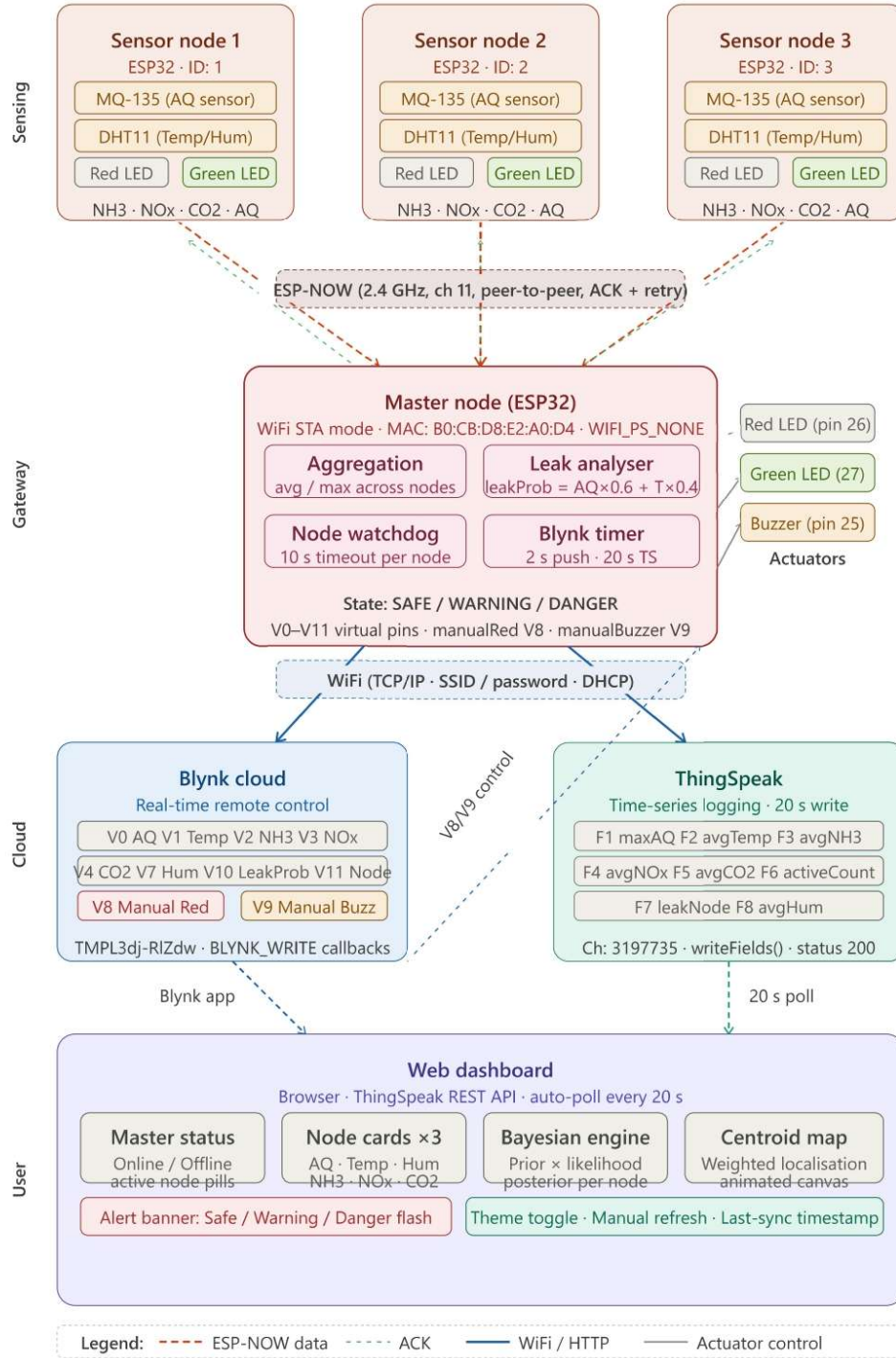
This module is responsible for sending data to cloud platforms for storage and visualization. It allows users to monitor system parameters such as gas levels, temperature, humidity, and leak probability through dashboards. This enables remote access and analysis of system performance.

4.2.8. Power Management Module

This module ensures proper power supply to all components of the system. It manages energy consumption of sensor nodes and maintains stable operation of the system. Efficient power management is important for reliable and continuous monitoring.

5. Methodology

5.1. System Design



5.2. Steps Involved in System Design and Implementation.

Steps Involved in System Design and Implementation

The development of the proposed system involves a series of systematic steps to ensure proper design, implementation, and testing. Each step plays an important role in achieving a reliable and efficient safety monitoring system.

5.2.1. Problem Identification

The first step is to identify the problem of gas leakage detection in industrial environments. The limitations of existing systems such as delayed response, lack of real-time monitoring, and absence of localization are studied to define the project objectives.

5.2.2. Requirement Analysis

In this step, the system requirements are determined. This includes selecting suitable sensors for gas detection, temperature, and humidity measurement, choosing an appropriate microcontroller, and deciding the communication method for data transfer.

5.2.3. System Design

The overall system architecture is designed by dividing the system into sensor nodes and a master node. The placement of sensor nodes is planned to ensure proper coverage of the monitoring area. Data flow and communication between components are also defined.

5.2.4. Hardware Selection and Setup

Components such as ESP32 microcontrollers, gas sensors, temperature and humidity sensors, LEDs, and buzzers are selected and connected. Proper interfacing of sensors with the microcontroller is carried out to ensure accurate data collection.

5.2.5. Software Development

The software for both sensor nodes and the master node is developed. This includes writing code for sensor data acquisition, communication using ESP-NOW, and processing of data at the master node. Logic for alert generation and system status determination is also implemented.

5.2.6. Communication Implementation

Wireless communication between sensor nodes and the master node is established using ESP-NOW protocol. Data packets are designed and transmitted efficiently to ensure reliable and low-latency communication.

5.2.7. Data Processing and Analysis

The master node processes the received data to evaluate environmental conditions. A probabilistic approach is used to estimate the likelihood of gas leakage, and the system determines whether the condition is safe, warning, or dangerous.

5.2.8. Alert System Integration

An alert mechanism is implemented to notify users about hazardous conditions. LEDs and buzzers are used for local alerts, and cloud platforms are integrated for remote monitoring and notifications.

5.2.9. Testing and Validation

The system is tested under different conditions to verify its performance. Sensor readings, communication reliability, and alert responses are checked to ensure proper functioning of the system.

5.2.10. Result Analysis and Optimization

The results obtained from testing are analyzed to evaluate system performance. Necessary improvements are made to enhance accuracy, reliability, and efficiency of the system.

5.3. System Simulation.

The proposed system, “**An IoT-Based Factory Safety System for Gas Leak Localization Using a Bayesian–WCL Hybrid Framework,**” was validated through a combination of **practical hardware implementation, real-time testing, and experimental analysis** rather than relying on software-based simulations. This approach ensures that the system performance reflects real-world industrial conditions and provides more reliable and accurate results.

5.3.1. System Design and Initial Architecture Development

The development of the system began with designing a complete architecture from scratch. A detailed system architecture was created to define the interaction between sensor nodes, communication modules, processing units, and cloud platforms. The architecture included multiple ESP32-based sensor nodes, a central master node, wireless communication using ESP-NOW, and integration with cloud services such as ThingSpeak and Blynk.

This design phase acted as a conceptual simulation, where different system components and their interactions were carefully planned before actual implementation. Decisions such as node placement, communication protocol selection, and data flow were finalized at this stage to ensure efficient real-time operation.

5.3.2. Hardware Implementation and Real-Time Testing

Unlike conventional approaches that depend on virtual simulation tools, this project was implemented entirely using **physical hardware components**. The complete circuit was built using ESP32 microcontrollers, MQ-135 gas sensors, DHT11 temperature and humidity sensors, LEDs, and buzzers.

Each sensor node was assembled and tested individually to verify its functionality. A multimeter was used to ensure proper voltage levels, continuity, and correct connections in the circuit. This step helped in identifying and eliminating hardware-level errors early in the development process.

After individual testing, all sensor nodes were integrated into the system and deployed in different positions to simulate a distributed industrial environment. Real-time data was collected from each node to verify sensor accuracy and responsiveness.

5.3.3. Software Development and Testing

The software for the system was developed using the Arduino IDE. Each ESP32 node was programmed to read sensor data, process it, and transmit it wirelessly using the ESP-NOW protocol.

The master node was programmed to:

- Receive data from all sensor nodes
- Aggregate sensor readings
- Apply the WCL algorithm for localization
- Perform Bayesian analysis for risk assessment
- Trigger alerts based on system state

The software was tested in multiple stages:

- **Unit Testing:** Individual sensor reading validation
- **Communication Testing:** Verifying ESP-NOW data transmission between nodes
- **Integration Testing:** Ensuring proper coordination between hardware and software

Debugging was performed using serial monitoring to analyze real-time data flow and identify logical or communication errors.

5.3.4. Communication Validation (ESP-NOW Protocol)

The ESP-NOW communication protocol was tested extensively to ensure reliable and low-latency data transfer. Multiple sensor nodes transmitted data simultaneously to the master node, and performance was evaluated based on:

- Data transmission speed
- Packet delivery reliability
- Communication range
- Delay between sensing and reception

The results confirmed that ESP-NOW provides **fast and efficient peer-to-peer communication**, making it suitable for real-time industrial applications.

5.3.5. Implementation of Weighted Centroid Localization (WCL)

In the Weighted Centroid Localization method, the estimated position of the gas leak is calculated using the weighted average of the positions of all sensor nodes. The weights are assigned based on the gas concentration detected by each node.

The mathematical representation is:

$$x = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i}, y = \frac{\sum_{i=1}^n w_i y_i}{\sum_{i=1}^n w_i}$$

➤ **Explanation of Terms**

- $x, y \rightarrow$ Estimated coordinates of gas leak
- $x_i, y_i \rightarrow$ Coordinates of the *i-th sensor node
- $w_i \rightarrow$ Weight of the node (based on gas concentration)
- $n \rightarrow$ Total number of sensor nodes

➤ **Weight Calculation in Your Project**

In your system:

- Weight is proportional to gas concentration (AQI)

So,

$$w_i = \text{Gas Concentration at node } i$$

Higher gas value $\rightarrow$ Higher weight $\rightarrow$ More influence

➤ **How it works in YOUR project**

- You have **3 nodes**
- Each node sends:
 - AQI
 - Temp
 - Gas values

Example:

Node	Gas Value	Weight
Node 1	120	High
Node 2	80	Medium
Node 3	40	Low

Node 1 pulls the location closer to itself

➤ **Why WCL is used**

- Simple and fast
- No complex hardware needed
- Works well for **small industrial areas**
- Suitable for ESP32 (low computation)

5.3.6. Implementation of Bayesian Theorem for Leak Probability Estimation

The proposed system uses a **Bayesian-inspired probabilistic approach** to estimate the likelihood of gas leakage based on sensor data. Unlike traditional threshold-based systems, this method evaluates multiple parameters to make a more reliable decision.

The mathematical expression of Bayes' theorem is:

$$P(L | D) = \frac{P(D | L) \cdot P(L)}{P(D)}$$

➤ Explanation of Terms

- $P(L | D)$ → Probability of leak given sensor data (**Posterior**)
- $P(L)$ → Prior probability of leak (initial assumption)
- $P(D | L)$ → Likelihood of observing data if leak exists
- $P(D)$ → Total probability of observed data

➤ How it is USED in This Project

In your system:

- **D (Data)** = Sensor readings
 - AQI (gas concentration)
 - Temperature
 - NH₃, CO₂
- The system evaluates:
 - If gas is high → higher chance of leak
 - If temperature is high → increases risk

These values are combined to estimate **P(L|D)**

➤ Practical Implementation

Your project uses a **simplified Bayesian-inspired model**, not full statistical training.

Example logic:

- Gas concentration → major factor
- Temperature → secondary factor

👉 Combined to calculate:

$$\text{Leak Probability} \approx f(\text{Gas, Temperature, Gases})$$

Result:

- Output in **percentage (0–100%)**

◆ **Why Bayesian is used**

- Handles uncertainty
- Reduces false alarms
- Better than fixed threshold
- Combines multiple sensor inputs

◆ **Example (easy understanding)**

Condition	Interpretation
Low gas + normal temp	Low probability
High gas + high temp	High probability
Medium gas	Warning

➤ **In your system classification**

Probability	Status
0–30%	Safe
30–60%	Warning
60–100%	Danger

5.3.7. Alert Mechanism

The alert mechanism is responsible for notifying users about the system condition based on the calculated leak probability. The system classifies the environment into three levels:

- **Safe Condition (0–30%):** Indicates normal operation with no immediate risk
- **Warning Condition (30–60%):** Indicates a potential risk that requires attention
- **Danger Condition (60–100%):** Indicates a high probability of gas leakage

In the proposed system, alerts are generated in two forms:

1. Local Alerts

The master node activates LEDs and a buzzer to provide immediate warning. A green LED indicates safe conditions, while a red LED and buzzer are activated during dangerous situations. This ensures that nearby personnel are alerted instantly.

2. Remote Monitoring

The system also transmits data to IoT platforms such as Blynk and ThingSpeak. These platforms display real-time values of gas concentration, temperature, humidity, and leak probability. Users can monitor system status remotely and take necessary action.

The alert mechanism continuously updates based on incoming sensor data, ensuring real-time response to changing conditions.

3. Cloud Integration and Remote Monitoring

The system was integrated with ThingSpeak and Blynk platforms to enable real-time data visualization and remote monitoring.

Testing included:

- Verifying data transmission from master node to cloud
- Monitoring real-time sensor values on dashboards
- Checking update intervals and data consistency
- Testing remote control features via Blynk

The cloud platforms successfully displayed real-time data, allowing users to monitor system status from anywhere. This confirmed the effectiveness of IoT integration.

5.3.8. Web Dashboard Development and Testing

A custom web dashboard was developed to provide a user-friendly interface for monitoring system data. The dashboard displayed:

- Sensor readings from all nodes
- System status (Safe/Warning/Danger)
- Leak probability values
- Localization results

The dashboard was tested for:

- Real-time data updates
- Responsiveness
- Accuracy of displayed values

This provided an additional layer of monitoring beyond mobile applications.

5.3.9. Experimental Testing in Different Environments

To validate system performance under real-world conditions, the system was tested in multiple environments. This included:

- Indoor controlled environments
- Open spaces with airflow variations
- Areas with different temperature and humidity conditions

Gas exposure was simulated using controlled sources, and system response was observed.

These experiments helped in understanding:

- Sensor behavior in different conditions
- Impact of environmental factors on localization
- Reliability of alert mechanisms

The system consistently detected gas presence, provided alerts, and estimated leak locations effectively.

5.3.10. Overall Validation and Performance Analysis

The complete system was evaluated based on:

- Response time
- Accuracy of detection
- Localization performance
- Reliability of communication
- Effectiveness of alerts

The results confirmed that the system successfully meets the objectives of:

- Real-time monitoring
- Gas leak detection
- Leak localization using WCL

Intelligent risk analysis using Bayesian methods

5.4. System Component Selection.

The selection of system components is a critical step in the design and implementation of the proposed **IoT-Based Factory Safety System for Gas Leak Localization Using a Bayesian–WCL Hybrid Framework**. The components were chosen based on the system’s input/output requirements, real-time performance needs, cost-effectiveness, reliability, and ease of integration. The system is divided into two main parts: **Sensor Nodes** and the **Master Node**, each with specific hardware requirements

5.5. Input and Output Requirements.

5.5.1. Input Requirements

The system requires continuous monitoring of environmental parameters to detect hazardous conditions. The key inputs include:

- **Gas Concentration:** Detection of harmful gases such as NH_3 , CO_2 , and other pollutants
- **Temperature and Humidity:** Environmental conditions affecting gas spread and sensor behavior
- **Flame Detection:** Additional safety parameter for fire hazards

These inputs must be captured accurately and in real time from multiple locations within the industrial environment.

5.5.2. Output Requirements

The system must provide both **local and remote outputs** for effective safety response:

- **Visual Alerts:** LEDs indicating Safe or Danger conditions
- **Audio Alerts:** Buzzer for immediate warning
- **Data Transmission:** Wireless communication to the master node
- **Cloud Output:** Real-time data visualization via ThingSpeak and Blynk

The outputs should be fast, reliable, and easily interpretable by users.

5.6. Sensor Node Component Selection

Each sensor node is responsible for sensing and transmitting environmental data. Three identical sensor nodes are deployed to ensure distributed monitoring.

5.6.1. ESP32 Microcontroller

The ESP32 was selected as the main controller due to the following reasons:

- Built-in **Wi-Fi and ESP-NOW support** for wireless communication
- High processing capability suitable for real-time applications
- Multiple GPIO pins for connecting sensors and actuators
- Low power consumption and cost-effectiveness
- Easy programming using Arduino IDE

The ESP32 acts as the core unit, handling sensor data acquisition and communication with the master node.



5.6.2. MQ-135 Gas Sensor

The MQ-135 sensor is used for detecting harmful gases and air quality levels.

Reasons for Selection:

- Capable of detecting multiple gases such as NH_3 , CO_2 , NO_x , and smoke

- Low cost and widely available
- Suitable for indoor industrial environments
- Analog output allows continuous monitoring of gas concentration



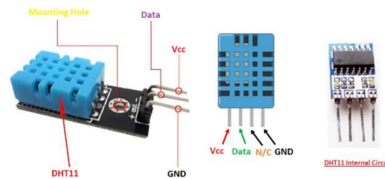
This sensor forms the primary input for detecting hazardous gas conditions.

5.6.3. DHT11 Temperature and Humidity Sensor

The DHT11 sensor is used to measure environmental temperature and humidity.

Reasons for Selection:

- Simple interface with digital output
- Low cost and easy integration with ESP32
- Provides sufficient accuracy for environmental monitoring
- Helps in improving analysis of gas behavior



Temperature and humidity data are important for enhancing the accuracy of gas detection and Bayesian analysis.

5.6.4. LEDs (Red and Green)

Each sensor node includes two LEDs for local indication:

- **Green LED:** Indicates safe conditions
- **Red LED:** Indicates presence of hazardous conditions



Reasons for Selection:

- Simple and effective visual indication
- Low power consumption
- Immediate feedback at sensor location

5.7. Master Node Component Selection.

The master node acts as the central processing unit of the system.

5.7.1. ESP32 Microcontroller (Master Node)

A separate ESP32 is used as the master node for:

- Receiving data from all sensor nodes via ESP-Now Protocol (peer to peer)
- Performing data aggregation
- Running WCL localization algorithm
- Executing Bayesian analysis



- Communicating with cloud platforms

Reasons for Selection:

- Supports both ESP-NOW and Wi-Fi simultaneously
- Sufficient processing power for handling multiple nodes
- Reliable and efficient for real-time systems

5.7.2. Buzzer

The buzzer is used as an audio alert system.



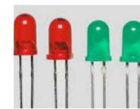
Reasons for Selection:

- Provides immediate audible warning during dangerous conditions
- Simple to interface with ESP32
- Low cost and effective for industrial environments

5.7.3. LEDs (Green and Red)

The master node also includes LEDs for centralized status indication:

- **Green LED:** Indicates overall system is safe
- **Red LED:** Indicates danger or gas leak detected



These provide quick visual feedback about system status.

5.8. Hardware Selection Procedure.

The selection of hardware components was carried out based on the following criteria:

5.8.1. Performance Requirements

Components were selected to ensure real-time data acquisition, processing, and communication. The ESP32 was chosen for its high performance and integrated communication capabilities.

5.8.2. Cost-Effectiveness

Low-cost components such as MQ-135 and DHT11 were selected to make the system affordable and scalable for industrial deployment.

5.8.3. Compatibility and Integration

All components were selected to ensure easy interfacing with the ESP32 microcontroller and compatibility with Arduino IDE for programming.

5.8.4. Reliability and Availability

Widely used and easily available components were preferred to ensure reliability and ease of replacement.

5.8.5. Scalability

The system is designed to allow addition of more sensor nodes without major hardware changes, making it scalable for larger environments.

6. Implementation

The implementation of the proposed IoT-based safety monitoring and alert system is carried out in a systematic and phased manner to ensure proper development, testing, and deployment. The system integrates hardware components, wireless communication, data processing, and a web-based dashboard for real-time monitoring and visualization.

6.1.Implementation Strategy

The system is implemented using a distributed architecture consisting of multiple sensor nodes and a central master node. Sensor nodes collect environmental data and transmit it to the master node using ESP-NOW communication. The master node processes the data, performs analysis, and sends the results to IoT platforms and a web dashboard for visualization.

6.2.Hardware Implementation Using ESP32 and Sensors.

The hardware implementation of the proposed system is based on a distributed architecture consisting of multiple sensor nodes and a central master node. The system utilizes ESP32 microcontrollers along with gas and environmental sensors to monitor industrial conditions in real time. The design focuses on reliability, low cost, and ease of deployment in industrial environments.

The sensor nodes are built using the ESP32 microcontroller, which acts as the core processing and communication unit. The ESP32 is selected due to its high processing capability, built-in Wi-Fi support, multiple GPIO pins, and compatibility with low-power applications. It also supports ESP-NOW communication, which enables fast and reliable data transfer between nodes without requiring internet connectivity.

Each sensor node is equipped with an MQ-135 gas sensor for detecting harmful gases such as ammonia (NH_3), carbon dioxide (CO_2), and nitrogen oxides (NO_x). The MQ-135 sensor operates by changing its internal resistance based on the concentration of gases present in the air. This change is converted into an analog voltage, which is read by the ESP32 through its ADC pins. The sensor provides continuous monitoring of air quality and helps in detecting abnormal gas levels.

In addition to gas sensing, a DHT11 sensor is used to measure temperature and humidity. Environmental conditions such as temperature and humidity can influence gas behavior and sensor readings. Therefore, incorporating these parameters improves the accuracy and reliability of the system. The DHT11 sensor provides digital output, which is directly read by the ESP32.

The sensor node also includes LED indicators to display local status. These indicators provide a quick visual representation of system conditions, such as normal or hazardous states.

The master node is also implemented using an ESP32 microcontroller. It acts as the central processing unit of the system. The master node receives data from all sensor nodes via ESP-NOW communication. It processes the received data, calculates leak probability, and determines the system status.

To provide alerts, the master node is equipped with LEDs and a buzzer. The LEDs indicate different safety levels, while the buzzer provides an audible alert in case of dangerous conditions. This ensures immediate notification to nearby personnel.

Proper power supply is provided to all components to ensure stable operation. The ESP32 operates at 3.3V, while sensors are powered according to their specifications. Voltage regulation and proper wiring are maintained to avoid fluctuations and ensure reliable performance.

The hardware components are connected using appropriate interfacing techniques, ensuring compatibility between sensors and microcontrollers. The modular design of the system allows easy expansion by adding more sensor nodes if required.

In conclusion, the hardware implementation of the system integrates ESP32 microcontrollers with gas and environmental sensors to create a reliable and efficient monitoring system. The use of distributed sensor nodes and a central master node enhances system coverage and enables real-time detection and alert generation.

6.3. Software Implementation (ESP-NOW, Blynk, ThingSpeak, Web Dashboard)

The software implementation of the proposed system plays a crucial role in enabling data acquisition, wireless communication, data processing, and real-time monitoring. The system software is developed for both sensor nodes and the master node using embedded programming on the ESP32 platform. In addition, IoT platforms such as Blynk, ThingSpeak, and a custom web dashboard are integrated for visualization and remote monitoring.

6.3.1. ESP32 Programming and Data Acquisition

The sensor nodes are programmed using the ESP32 microcontroller to continuously read data from the MQ-135 gas sensor and DHT11 temperature-humidity sensor. The analog output from the gas sensor is converted into digital values using the ADC of the ESP32, while the DHT11 provides digital readings directly.

The software performs basic preprocessing such as filtering and formatting of sensor data before transmission. Each sensor node operates independently and periodically sends updated data to the master node.

6.3.2. ESP-NOW Communication Protocol

ESP-NOW is used as the primary communication protocol between sensor nodes and the master node. It is a low-power, peer-to-peer wireless communication protocol supported by ESP32 devices.

In the software implementation, each ESP32 device is configured with a unique MAC address, and peer devices are registered for communication. Sensor data is packed into structured data frames and transmitted wirelessly to the master node.

ESP-NOW offers several advantages:

- Low latency communication
- No dependency on internet connectivity
- Reliable transmission in short-range industrial environments

The master node continuously listens for incoming data packets and processes them in real time.

6.3.3. Data Processing and Decision Logic

The master node software is responsible for processing all incoming data from sensor nodes. It performs the following operations:

- Aggregation of sensor data from multiple nodes
- Calculation of gas leak probability using a Bayesian-inspired approach
- Estimation of leak location using Weighted Centroid Localization
- Classification of system status into safe, warning, or danger

The processed results are then used for both local alerts and cloud updates.

6.3.4. Blynk Integration

Blynk is used as a mobile-based IoT platform for real-time monitoring and user interaction. The ESP32 master node sends processed data to the Blynk server using Wi-Fi connectivity.

The Blynk dashboard displays:

- Gas concentration values
- Temperature and humidity
- Leak probability
- System status indicators

Users can monitor the system remotely through the Blynk mobile application. The platform provides a user-friendly interface and allows real-time updates.

6.3.5. ThingSpeak Integration

ThingSpeak is used for data logging and graphical visualization. The master node sends sensor data and calculated values to the ThingSpeak cloud platform at regular intervals.

ThingSpeak provides:

- Time-based data storage
- Graphical representation of sensor readings
- Trend analysis of gas concentration and environmental parameters

This helps in analyzing system performance and observing patterns over time.

6.3.6. Web Dashboard Implementation

In addition to IoT platforms, a custom web dashboard is developed for enhanced visualization and monitoring. The web dashboard is designed to display real-time system data in a structured and interactive manner.

The dashboard includes:

- Live sensor readings from all nodes
- Leak probability visualization
- System status indicators (Safe/Warning/Danger)
- Graphical charts and data trends
- Node-wise monitoring information

The web dashboard receives data from the master node either directly or through cloud integration. It provides a centralized interface for monitoring the entire system and improves user accessibility.

6.3.7. Alert System Integration

The software also includes logic for generating alerts based on system conditions. When the calculated probability exceeds predefined thresholds, the system activates:

- LEDs for visual indication
- Buzzer for audible alerts
- Notifications on IoT platforms

This ensures immediate response to hazardous situations.

6.4. Development Phases of the Project

6.4.1. Phase 1: Requirement Analysis and Planning

In this phase, the problem of gas leakage detection and industrial safety monitoring is analyzed. Requirements such as real-time monitoring, wireless communication, localization, and remote visualization are identified. Components such as ESP32, MQ-135, and DHT11 are selected.

The need for a **web dashboard** is also identified to enable remote monitoring and visualization of system data.

6.4.2. Phase 2: Hardware Setup and Circuit Design

Sensor nodes are assembled using ESP32 microcontrollers connected to gas sensors and temperature-humidity sensors. LEDs are included for local indication.

The master node is developed using ESP32, along with LEDs and a buzzer for alert generation. Proper power supply and connections are ensured.

6.4.3. Phase 3: Software Development

Software is developed for both sensor nodes and the master node.

- Sensor nodes read environmental data and transmit it using ESP-NOW
- Master node receives data and performs:
 - Gas analysis
 - Leak probability calculation
 - Localization using WCL

The system logic for classification and alert generation is implemented.

6.4.4. Phase 4: Communication Setup (ESP-NOW Integration)

ESP-NOW protocol is configured for communication between nodes. Each node is assigned a unique ID, and peer communication is established.

Sensor data is transmitted in structured packets and received by the master node in real time.

6.4.5. Phase 5: Data Processing and Alert Mechanism

The master node processes incoming data and calculates leak probability using a Bayesian-inspired approach. Based on the result, the system determines the safety level.

Alerts are generated as:

- LED indicators
- Buzzer alerts
- Data transmission to cloud platforms

6.4.6. Phase 6: Web Dashboard and Cloud Integration

A web-based dashboard is developed to visualize system data in real time. The master node sends processed data to IoT platforms such as Blynk and ThingSpeak, which are used to display system parameters.

The web dashboard displays:

- Gas concentration levels
- Temperature and humidity

- Leak probability (%)
- Node status (active/inactive)
- System condition (Safe/Warning/Danger)

The dashboard may include graphical representations such as charts and indicators to help users easily understand system behavior.

This module enables:

- Remote monitoring
- Real-time visualization
- Data logging and analysis

6.4.7. Phase 7: Testing and Validation

The complete system, including the web dashboard, is tested under different conditions. Sensor readings, communication reliability, dashboard updates, and alert responses are verified.

The system is checked for real-time performance and accuracy.

6.4.8. Phase 8: Optimization and Final Deployment

The system is optimized by calibrating sensors, improving communication efficiency, and refining dashboard display. Threshold values and probability calculations are adjusted for better accuracy.

The final system is deployed in a controlled environment for demonstration.

7. Result and Discussion

7.1. Hardware Results and Sensor Data Analysis.

The hardware implementation of the proposed IoT-based safety monitoring system was successfully developed using ESP32-based sensor nodes and a central master node. The system was enclosed in custom-designed 3D-printed casings, providing a compact, robust, and sophisticated structure suitable for industrial environments.



7.1 MASTER NODE

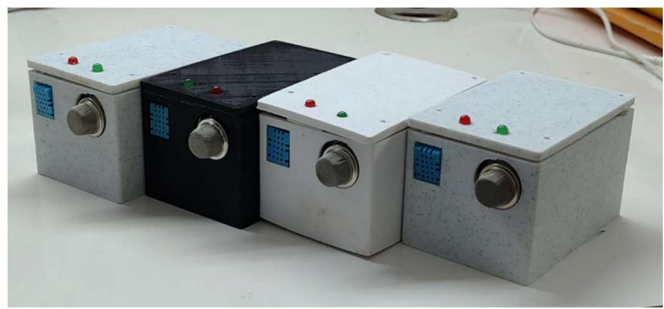


FIG 7.2 Sensor Node

The complete hardware setup consists of a master node and multiple sensor nodes, as shown in **Fig. 7.1** and **Fig. 7.2**. The master node (Fig. 7.1) includes an ESP32 microcontroller, LED indicators, and a buzzer, which together perform data processing and alert generation. The sensor nodes (Fig. 7.2) are equipped with MQ-135 gas sensors and DHT11 temperature-humidity sensors, enabling continuous monitoring of environmental conditions.



Fig 7.3 Overall System

The placement of sensor nodes within the monitoring area, as illustrated in **Fig. 7.3**, allows for distributed sensing and effective spatial coverage. This arrangement enables the system to detect variations in gas concentration across different locations.

During experimental evaluation, the hardware system demonstrated reliable performance. The MQ-135 gas sensors showed high sensitivity to changes in gas concentration, with noticeable increases in output values when exposed to higher gas levels. The DHT11 sensors provided stable temperature readings in the range of approximately 27°C to 30°C, indicating proper sensor calibration and environmental consistency.

The collected data from multiple sensor nodes enabled comparative analysis of gas concentration levels. It was observed that the sensor node closest to the gas source consistently recorded higher values, while nodes farther away recorded comparatively lower values. This confirms the effectiveness of the distributed sensing approach.

The system further processed the collected data to compute normalized gas intensity for each node. The node with the highest gas concentration exhibited the highest normalized intensity and corresponding leak probability. This behavior validates the effectiveness of the Bayesian-inspired probability model.

Additionally, the Weighted Centroid Localization (WCL) method was applied to estimate the leak location. The results showed that the calculated position shifted toward the node with the highest gas concentration, confirming accurate localization of the leak region.

The hardware system also demonstrated fast response time, with immediate activation of LED indicators and buzzer during hazardous conditions. This ensures rapid detection and timely alert generation.

Overall, the hardware results confirm that the system successfully performs gas detection, environmental monitoring, probabilistic analysis, and leak localization with high reliability.

7.2. Software Results and Dashboard Visualization.

The software implementation of the system was evaluated using a custom web dashboard along with Blynk and ThingSpeak platforms. The system successfully demonstrated real-time data processing, visualization, and remote monitoring capabilities.

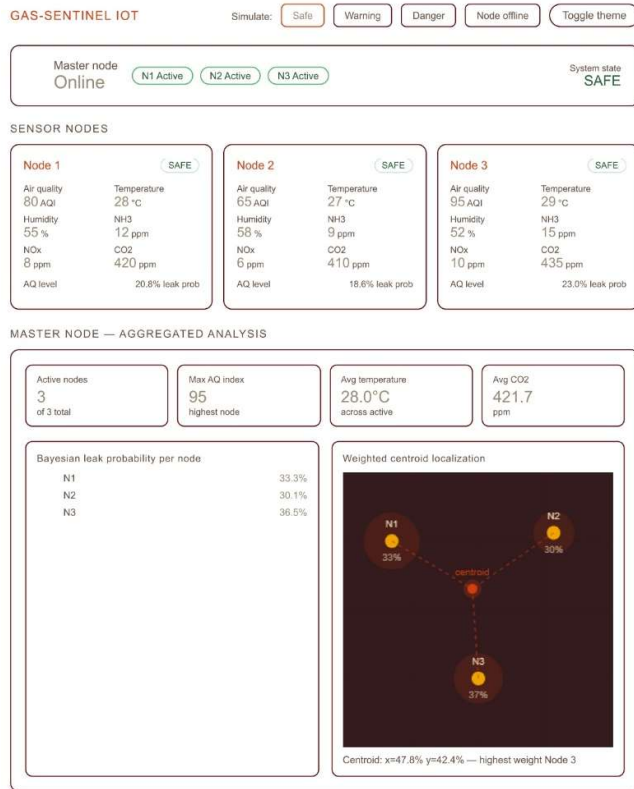


Fig 7.4 Dashboard – Safe Condition

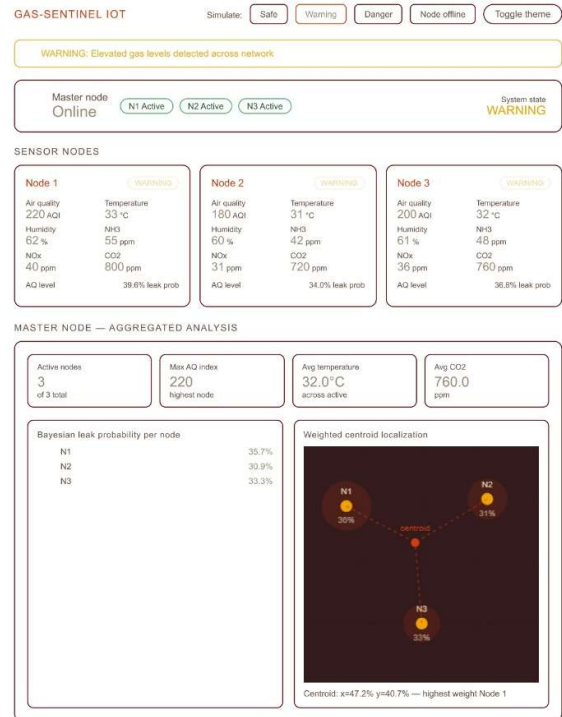


Fig 7.5 Dashboard – Warning Condition



Fig 7.6 Dashboard – Danger Condition



Fig 7.7 Dashboard – Node Offline Scenario

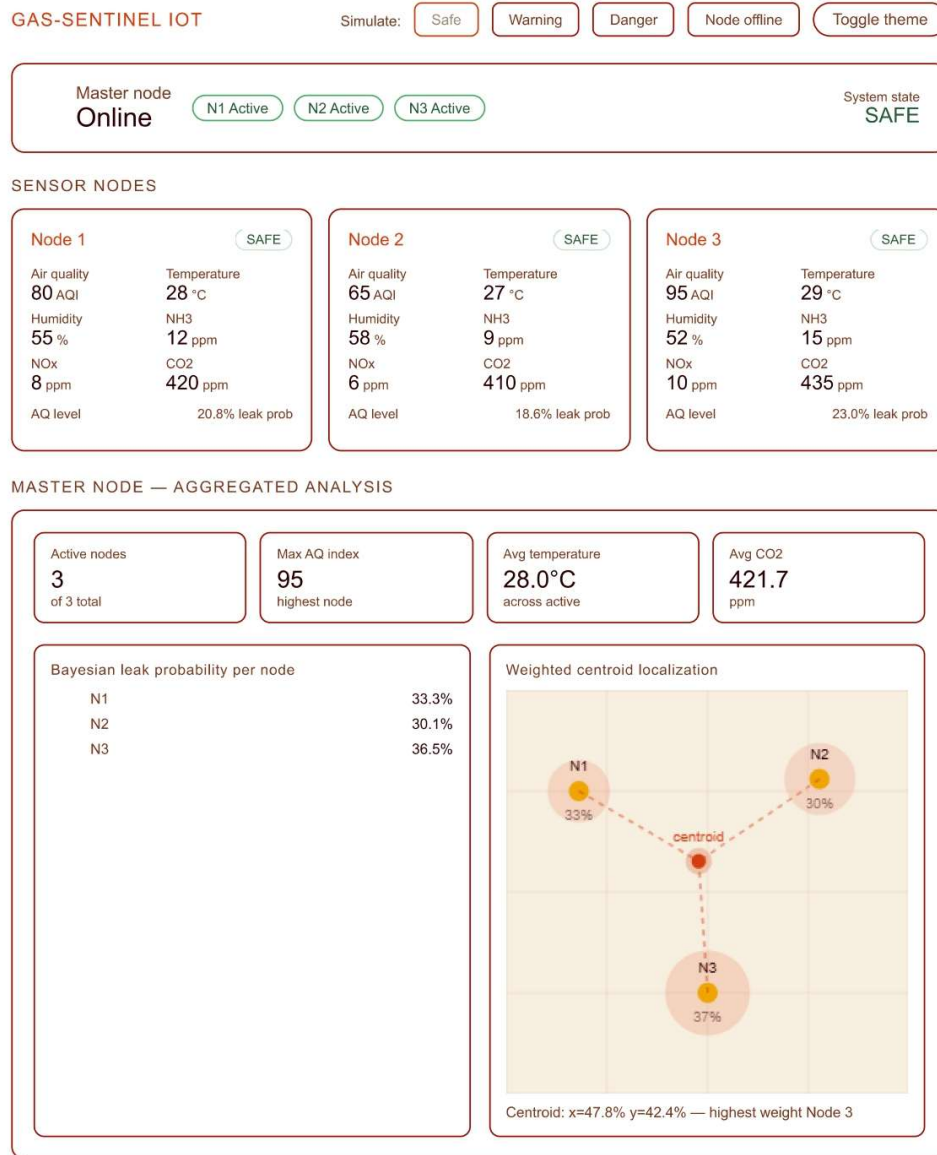


Fig 7.8 Dashboard – Light Theme Interface

The web dashboard interface, shown in **Fig. 7.4 to Fig. 7.8**, provides a comprehensive visualization of system parameters, including gas concentration, temperature, humidity, leak probability, and node status. The dashboard retrieves data in real time using the ThingSpeak API, ensuring seamless integration between the hardware and cloud platforms.

7.2.1. Safe Condition

As shown in **Fig. 7.4**, under normal environmental conditions, all sensor nodes report low gas concentration values. The leak probability remains low, and the system is classified as Safe.

The dashboard displays stable readings, confirming proper system operation without false alarms.

7.2.2. Warning Condition

In **Fig. 7.5**, moderate gas concentration levels are observed. The system detects an increase in gas values and updates the leak probability accordingly. The system transitions into a Warning state, indicating a potential risk that requires attention.

7.2.3. Danger Condition

As illustrated in **Fig. 7.6**, high gas concentration levels result in a significant increase in leak probability. The system classifies this condition as Danger. The node closest to the gas source records the highest values, validating the effectiveness of Bayesian analysis and localization.

7.2.4. Node Offline Scenario

In **Fig. 7.7**, one of the sensor nodes becomes inactive. The dashboard accurately reflects the node offline status while continuing to process data from the remaining active nodes. The localization algorithm adapts dynamically, ensuring uninterrupted system operation.

7.2.5. Dashboard Features and User Interface

The web dashboard also includes advanced features such as real-time updates, graphical visualization, and a theme toggle option, as shown in **Fig. 7.8**. The ability to switch between light and dark modes enhances usability and user experience.

In addition to the web dashboard, the Blynk platform provides mobile-based monitoring, allowing users to access system data remotely. ThingSpeak is used for data logging and graphical trend analysis, enabling long-term observation of system behavior.

7.3. Overall Software Performance

7.3.1. Experimental Results

min	Node 1			Node 2			Node 3		
	CO ₂	NH ₃	smoke	CO ₂	NH ₃	Smoke	CO ₂	NH ₃	Smoke
0	120	35	60	115	30	55	118	32	58
5	130	38	65	125	34	60	128	36	62
10	150	42	70	140	38	65	145	40	68
15	220	60	95	180	50	80	200	55	90
20	300	85	130	210	60	95	260	75	115
25	400	110	170	250	75	110	320	95	145

30	520	140	210	300	90	130	400	120	180
35	600	160	240	350	100	150	450	135	210
40	650	180	260	400	115	170	500	150	230
45	700	200	280	450	130	190	550	165	250
50	720	210	300	500	140	210	580	175	270
55	750	230	320	520	150	220	600	185	290

Table 7.3.1. Experimental Results of Three-Node Gas Leakage Detection

This establishes a concentration gradient (Node 1 > Node 3 > Node 2), which forms the basis for subsequent Bayesian analysis and WCL-based localization. This is followed by Bayesian inference and subsequent WCL-based localization on the nodes, which results in a concentration gradient (Node 1 > Node 3 > Node 2), the justification for which is established through the collection of a multi-gas (CO₂, NH₃, and Smoke) time-series data set for 60 minutes from the three spatially distributed sensor nodes. To get a concise picture of the system behavior, the data is represented by the maximum value of the combined gas concentration at each node.

The total gas concentration for each node is computed as:

$$C_{total} = CO_2 + NH_3 + Smoke$$

From the dataset, the peak values are:

- Node 1 → 1300 ppm
- Node 2 → 890 ppm
- Node 3 → 1075 ppm

To enable comparison across nodes, these values are **normalized with respect to the global maximum (1300 ppm)**, yielding the **maximum gas intensity (%)**.

The normalized intensity is given by:

$$\text{Intensity (\%)} = \frac{C_{\text{node}}}{C_{\text{max}}} \times 100$$

This normalization ensures that all sensor readings are scaled consistently, allowing meaningful probabilistic interpretation.

Subsequently, a simplified **Bayesian inference model** is applied, where the normalized intensity is directly mapped to the **posterior probability of leak occurrence**, i.e.,

$$P(\text{Leak}) \propto \text{Normalized Intensity}$$

Based on the computed probabilities, the system classifies the risk levels into three categories:

Safe (0–0.3)

Warning (0.3–0.7)

Danger (>0.7)

Finally, the proximity to the leak source is inferred based on the generalized trend observed in Weighted Centroid Localization (WCL), where nodes with relatively higher gas concentration levels indicate closer proximity to the leak source. Therefore, Node 1 exhibits the highest amount of gas concentration and highest leak probability, proving its close proximity to the leak source” (There is a similarity or consistency between normalized gas intensity, Bayesian inference, and WCL-based localization, which proves that the proposed system works well.

7.3.2. Average Node Wise Data

Node ID	Max Gas Intensity (%)	Bayesian P(Leak)	Classification	Proximity to Source
Node 1	100	1.00	Danger	Nearest (~0.8 m)
Node 2	68.5	0.69	Warning	Far (~2.5 m)
Node 3	82.7	0.83	Danger	Medium (~1.5 m)

Table 7.3.2. Node-wise Maximum Gas Intensity and Bayesian Leak Probability-Based Risk Classification

7.3.3. Bayesian Inference Performance

The Bayesian posterior probability scores spatial gas concentration gradient implied across the three sensor nodes.

From the experimental dataset, Node 1 registers the highest normalized gas intensity (100%) that also produces a posterior leak probability of $P = 1.00$, which means that it is in the Danger regime.

Again, node 3, with a relatively high intensity of 82.7%, resulted in a postulated probability of $P = 0.83$, also falling within Danger but with relatively less confidence than node 1.

In contrast, Node 2, which registers a lower normalized intensity at 68.5%, yields a posterior probability value of $P = 0.69$, which falls within the Warning category, indicating moderate exposure.

The Bayesian formulation also has the benefit of providing a graded probabilistic interpretation of the severity of leaks as opposed to conventional fixed-threshold approaches. For instance, while Danger is a common category for Node 1 and Node 3, the posterior probabilities of 1.00 vs 0.83 help differentiate the severity of the Danger and the distance to the leak source, respectively.

Again, Node 2, with a probability of 0.69, presents a threshold state that cannot be clearly resolved with a dual threshold.

This probabilistic scaling allows a wider dynamic range of interpretation, so that different delicate scaling in decision-making vis-à-vis severe, moderate, and marginal leak conditions may also be distinguished with reliability for industrial safety applications.

7.3.4. Weighted Centroid Localization Performance

Finally, using the posterior probabilities obtained in Table.2 as weights with the given coordinates for sensor nodes, the Weighted Centroid Localisation, (WCL) method was used to estimate the position of a leak source.

Let:

- $P_1 = 1.00, P_2 = 0.69, P_3 = 0.83$
- Node coordinates:
 - Node 1 $\rightarrow (2, 3)$
 - Node 2 $\rightarrow (8, 3)$
 - Node 3 $\rightarrow (5, 9)$

Computation:

$$X = \frac{(1.00 \times 2) + (0.69 \times 8) + (0.83 \times 5)}{1.00 + 0.69 + 0.83} = \frac{2 + 5.52 + 4.15}{2.52} = \frac{11.67}{2.52} \approx 4.63$$

$$Y = \frac{(1.00 \times 3) + (0.69 \times 3) + (0.83 \times 9)}{2.52} = \frac{3 + 2.07 + 7.47}{2.52} = \frac{12.54}{2.52} \approx 4.98$$

Estimated Leak Location:

$$(4.63, 4.98)$$

Finally, the centroid position is recalculated, again shifting towards node 1, which recorded the highest gas level, indicating that the leak is closest to node 1. On the contrary, the contribution of node 3 is more likely to bias the estimate towards the upper region, whereas node 2 has a

minimal contribution due to its lower probability. This proves that WCL is able to incorporate multi-node probabilistic weights to provide a reliable estimation of the leak zone. Therefore, the localization result can be considered sufficient for a zone-level identification approach that allows the emergency response teams to know which zone in the industrial environment to focus on.

The software system demonstrated:

- Real-time data acquisition and visualization
- Accurate representation of system status
- Reliable communication with cloud platforms
- Seamless integration between hardware and software

The integration of Bayesian probability analysis and Weighted Centroid Localization provided meaningful insights into leak severity and location, which were effectively visualized through the dashboard.

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8. Conclusion and Future Scope

8.1. Conclusion of the Proposed System.

The proposed IoT-based safety monitoring and alert system for chemical factories has been successfully designed and implemented. The system effectively integrates distributed sensor nodes, wireless communication, probabilistic analysis, and real-time visualization to provide a comprehensive solution for gas leakage detection and monitoring.

The hardware implementation using ESP32 microcontrollers, MQ-135 gas sensors, and DHT11 temperature-humidity sensors demonstrated reliable and accurate sensing of environmental parameters. The use of custom 3D-printed enclosures enhanced the structural robustness and practical usability of the system in real-world scenarios.

The communication between sensor nodes and the master node using ESP-NOW protocol ensured fast, low-latency, and reliable data transmission without dependence on internet connectivity. The system was able to continuously collect and process data from multiple locations, enabling effective monitoring of the environment.

The integration of a Bayesian-inspired probabilistic model allowed the system to estimate leak probability in a more intelligent and adaptive manner compared to conventional threshold-based methods. This approach reduced false alarms and provided a graded interpretation of risk levels, categorized as Safe, Warning, and Danger.

Furthermore, the implementation of the Weighted Centroid Localization (WCL) technique enabled the system to estimate the approximate location of the gas leak. The localization results consistently indicated the region closest to the leak source, thereby improving response efficiency and safety.

The software implementation, including the web dashboard, Blynk, and ThingSpeak platforms, successfully provided real-time visualization and remote monitoring capabilities. The dashboard effectively displayed system parameters such as gas concentration, temperature, humidity, leak probability, and node status. The ability to monitor the system remotely enhances its practicality for industrial applications.

The system also demonstrated robustness under different scenarios, including safe conditions, warning levels, dangerous situations, and node failure cases. The ability to adapt to node offline conditions and continue operation highlights the fault tolerance of the system.

In conclusion, the proposed system offers a reliable, cost-effective, and scalable solution for industrial safety monitoring. It not only detects gas leakage but also provides intelligent analysis and localization, enabling timely and informed decision-making. The integration of IoT, probabilistic analysis, and real-time visualization makes the system highly suitable for deployment in chemical industries and other hazardous environments.

8.2.Future Scope and Enhancements.

The proposed IoT-based safety monitoring and alert system demonstrates effective gas detection, probabilistic analysis, and localization capabilities. However, there are several opportunities for further improvement and expansion to enhance system performance, scalability, and applicability in real-world industrial environments.

One of the primary areas for enhancement is the inclusion of **additional and more advanced gas sensors**. While the current system uses MQ-135 sensors for general air quality detection, integrating specialized sensors for specific gases such as methane (CH₄), carbon monoxide (CO), and hydrogen sulfide (H₂S) can improve detection accuracy and reliability in industrial applications.

The system can also be enhanced by implementing **advanced machine learning algorithms** for predictive analysis. By training models on historical sensor data, the system can predict potential gas leak events before they occur, enabling preventive maintenance and reducing the risk of accidents.

Another important improvement is the use of **more precise localization techniques**. While Weighted Centroid Localization provides a practical approximation, advanced methods such as trilateration, RSSI-based positioning, or hybrid localization techniques can be implemented to improve accuracy in larger industrial setups.

The communication system can be extended to support **long-range communication technologies** such as LoRa or NB-IoT. This would allow the system to be deployed in large-scale industrial environments where wide-area coverage is required.

The web dashboard can be further enhanced by adding **advanced visualization features**, such as real-time heat maps for gas concentration, historical trend analysis, and automated report generation. Integration with mobile notifications and alert systems can also improve user responsiveness.

Another potential enhancement is the integration of **automated safety mechanisms**. For example, the system can be connected to ventilation systems, exhaust fans, or emergency shutdown controls, enabling automatic response when dangerous conditions are detected.

Power management can also be improved by incorporating **low-power design techniques and energy-efficient components**, allowing the system to operate for longer durations, especially in remote or battery-powered deployments.

The system can be extended to support **cloud-based analytics and data storage**, enabling centralized monitoring of multiple industrial sites. This would make the system scalable and suitable for large organizations.

Finally, the integration of **security features** such as data encryption, secure communication protocols, and authentication mechanisms is essential to protect the system from unauthorized access and cyber threats.

Achievements of the Project

- The project titled “IoT-Based Safety Monitoring and Alert System for Chemical Factories” was successfully presented at the 14th International Conference on Contemporary Engineering and Technology (ICCET 2026) held in Chennai, India.
- The research work focused on a Bayesian-WCL hybrid framework for gas leak detection and localization, demonstrating real-time industrial safety applications.
- The project team also participated in the WIREX Hardware Competition, organized by the IETE Students Chapter, showcasing the hardware implementation and IoT integration of the system.
- The project received appreciation for:
 - Real-time monitoring capability
 - Accurate gas leak detection
 - Smart alert mechanism using IoT

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